

```
import pandas as pd
import matplotlib as mpt
import numpy as np
df=pd.read_csv("/content/Task_8_Cleaned_Retail_Sales.csv")
```

```
#Step 2 – Select numeric variables
# Select numeric columns for correlation analysis
numeric_df = df.select_dtypes(include='number')

print("Numeric columns:")
print(numeric_df.columns.tolist())

print("\nShape of numeric dataset:", numeric_df.shape)
```

```
Numeric columns:
['quantity', 'unit_price', 'discount_pct', 'gross_sales', 'discount_amount', 'net_sales']

Shape of numeric dataset: (120, 6)
```

```
#Step 3 – Calculate the Correlation Matrix
# Calculate Pearson correlation matrix
correlation_matrix = numeric_df.corr()

# Display the correlation matrix
correlation_matrix
```

	quantity	unit_price	discount_pct	gross_sales	discount_amount	net_sales	
quantity	1.000000	-0.030070	-0.116160	0.624278	0.302238	0.619867	
unit_price	-0.030070	1.000000	-0.069390	0.680383	0.347864	0.672669	
discount_pct	-0.116160	-0.069390	1.000000	-0.137345	0.628900	-0.245828	
gross_sales	0.624278	0.680383	-0.137345	1.000000	0.498648	0.990649	
discount_amount	0.302238	0.347864	0.628900	0.498648	1.000000	0.375725	
net_sales	0.619867	0.672669	-0.245828	0.990649	0.375725	1.000000	

Next steps: [Generate code with correlation_matrix](#) [New interactive sheet](#)

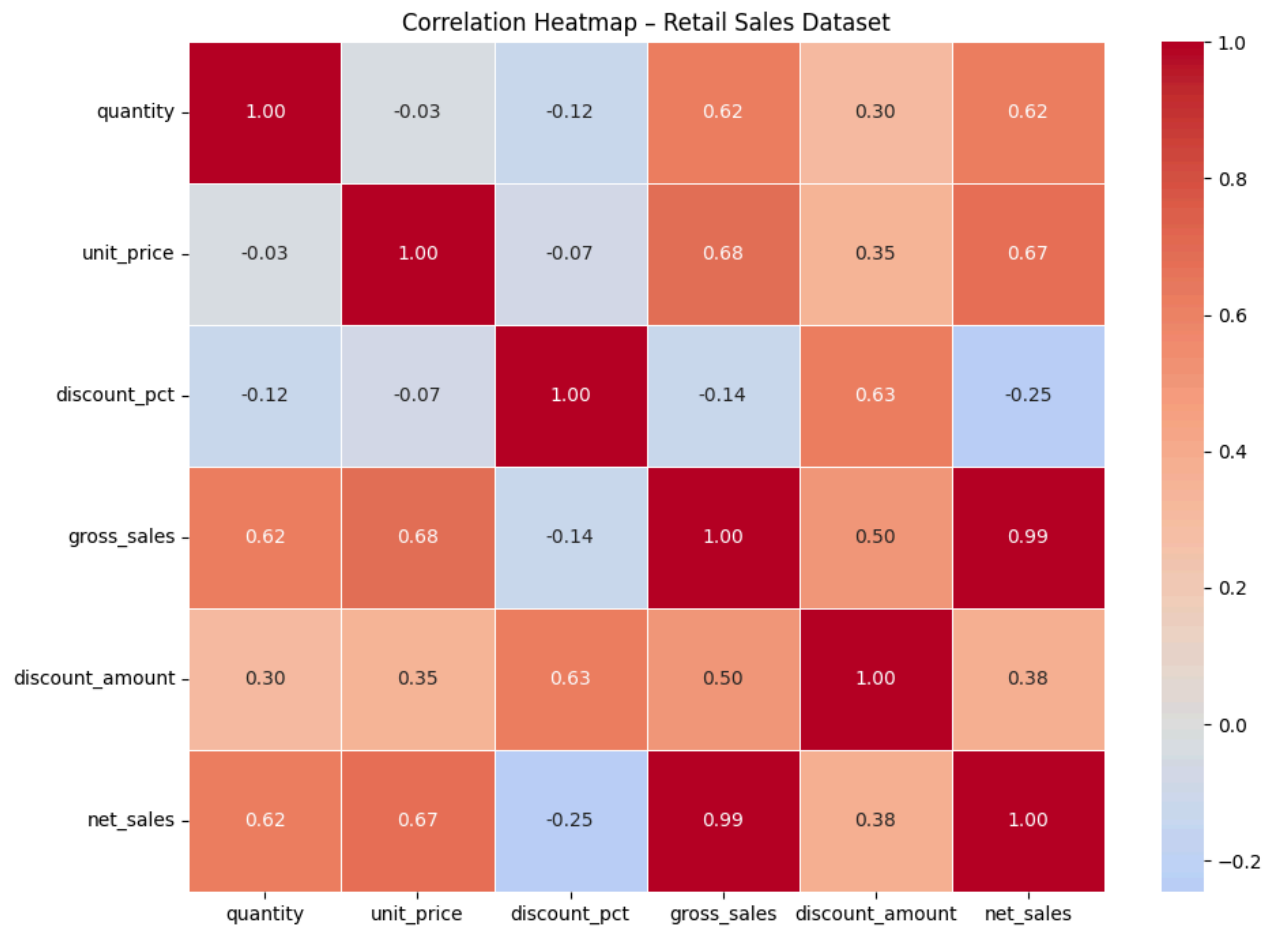
```
#Step 4 – Create the Heatmap
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5,
    center=0
)

plt.title("Correlation Heatmap - Retail Sales Dataset")
plt.tight_layout()

plt.show()
```



```

· Identify the strongest relationships
· unique correlation pairs
s = correlation_matrix.where(
    tril(np.ones(correlation_matrix.shape)).astype(bool)
).sort_values(ascending=False)

(Strongest correlation pairs:")
r_pairs)
I give us the relationships in descending order and make it easy to identify the strongest correlations

```

```

Strongest correlation pairs:
gross_sales    net_sales    0.990649
unit_price     gross_sales    0.680383
               net_sales     0.672669
discount_pct   discount_amount 0.628900
quantity       gross_sales    0.624278
               net_sales     0.619867
gross_sales    discount_amount 0.498648
discount_amount net_sales     0.375725
unit_price     discount_amount 0.347864
quantity       discount_amount 0.302238
               unit_price     -0.030070
unit_price     discount_pct   -0.069390
quantity       discount_pct   -0.116160
discount_pct   gross_sales    -0.137345
               net_sales     -0.245828

dtype: float64

```

Step 6 — Final interpretation

Based on your actual correlation matrix, these are the key relationships we should document:

Relationship	Correlation	Interpretation
Gross Sales ↔ Net Sales	0.991	Very strong positive
Unit Price ↔ Gross Sales	0.680	Moderate-to-strong positive
Unit Price ↔ Net Sales	0.673	Moderate-to-strong positive
Discount % ↔ Discount Amount	0.629	Moderate positive
Quantity ↔ Gross Sales	0.624	Moderate positive
Quantity ↔ Net Sales	0.620	Moderate positive
Discount % ↔ Net Sales	-0.246	Weak-to-moderate negative
Quantity ↔ Unit Price	-0.030	Almost no linear relationship

```
#mportant modelling finding

#The 0.991 correlation between Gross Sales and Net Sales indicates that these two variables contain al
#linear information. Including both as independent predictors in a model could create multicollinearit
```

```
#Step 7 – Save the heatmap as PNG
plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5,
    center=0
)

plt.title("Correlation Heatmap - Retail Sales Dataset")
plt.tight_layout()

plt.savefig("Task_11_Correlation_Heatmap.png", dpi=300, bbox_inches="tight")

plt.show()
```

